



APPENDIX A

Additional Derivations



Algebra's like sheet music. The important thing isn't can you read music, it's can you *hear* it. Can you hear the music, Robert?

— *Niels Bohr, in Oppenheimer* —



In this section, we present additional derivations for expressions used in Chapter 2.

A.1 | Roche lobe formula

We follow the derivation given in [Rose \(1998\)](#) to find an expression for the Roche potential. First we note that in a rotating reference frame in Cartesian coordinates, we have

$$\overbrace{\frac{GM_1M_2}{a^2}\hat{\mathbf{x}}}^{\text{gravitational force}} = \underbrace{-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1)}_{\text{centrifugal force}}, \quad (\text{A.1})$$

where $\hat{\mathbf{x}}$ is the line between the center of mass of both stars, a the separation, $\vec{\omega}$ the angular velocity, and $\vec{r}_1$ the position vector of the primary star. The x-ordinates of the centres of mass are given by

$$x_1 = \frac{M_2}{M_1 + M_2}a \quad (\text{A.2})$$

$$x_2 = \frac{-M_1}{M_1 + M_2}a \quad (\text{A.3})$$

Since we are considering a rotating coordinate system we have that the centrifugal force is given by

$$-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1) = M_1\omega^2x_1, \quad (\text{A.4})$$

and hence

$$\begin{aligned} \frac{GM_1M_2}{a^2}\hat{\mathbf{x}} &= M_1\omega^2x_1 \\ \omega^2 &= \frac{GM_1M_2}{M_1x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2}{x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2(M_1 + M_2)}{M_2 \cdot a \cdot a^2}\hat{\mathbf{x}} \\ &= \frac{G(M_1 + M_2)}{a^3}\hat{\mathbf{x}}. \end{aligned} \quad (\text{A.5})$$

The centrifugal potential of a test mass m at (x, y) is hence

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2) = -\frac{Gm(M_1 + M_2)(x^2 + y^2)}{2a^3}, \quad (\text{A.6})$$

and thus its energy is

$$E = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + V(x, y), \quad (\text{A.7})$$

where

$$V(x, y) = -\overbrace{\frac{GmM_1}{\sqrt{(x-x_1)^2 + y^2}}}^{\text{G.P.E. due to primary}} - \overbrace{\frac{GmM_2}{\sqrt{(x-x_2)^2 + y^2}}}^{\text{G.P.E. due to secondary}} - \overbrace{\frac{Gm(M_1 + M_2)(x^2 + y^2)}{2a^3}}^{\text{centrifugal potential}}. \quad (\text{A.8})$$

We introduce the dimensionless coordinates measured in units of the separation of the binary (or reduced masses)

$$\xi = \frac{x}{a}, \quad \eta = \frac{y}{a}, \quad \xi_1 = \frac{M_2}{M_1 + M_2}, \quad \xi_2 = \frac{M_1}{M_1 + M_2} \quad (\text{A.9})$$

Our potential now becomes

$$V(\xi, \eta) = -\frac{Gm(M_1 + M_2)}{a} \times \left[\frac{\xi_2}{\sqrt{(\xi - \xi_1)^2 + \eta^2}} + \frac{\xi_1}{\sqrt{(\xi - \xi_2)^2 + \eta^2}} + \frac{1}{2}(\xi^2 + \eta^2) \right]. \quad (\text{A.10})$$

Taking the partial derivative along the ξ -axis (the rescaled x -axis), we get

$$\frac{\partial V}{\partial \xi} = -\frac{Gm(M_1 + M_2)}{a} \times \left[-\frac{\xi_2(\xi - \xi_1)}{\left((\xi - \xi_1)^2 + \eta^2\right)^{\frac{3}{2}}} - \frac{\xi_1(\xi - \xi_2)}{\left((\xi - \xi_2)^2 + \eta^2\right)^{\frac{3}{2}}} + \xi \right] \quad (\text{A.11})$$

Hence the extrema of the potential V are at the points ξ where (taking $\eta = 0$ on the ξ -axis)

The formula for the Roche lobes of binary stars.

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}, \quad (\text{A.12})$$

which was to be shown.

A.2 | Kepler's Three Laws

This section is adapted from [Richards \(2021\)](#). Kepler's three laws such that they apply to binary stars are:

1. All binary stars orbit the barycentre in an ellipse with the barycentre as a focus.
2. A line segment joining the two stars sweeps out equal areas during equal intervals of time.
3. The square of the period of each star is proportional to the cube of the semimajor axis (i.e., $T^2 \propto a^3$).

The derivations are presented below.

A.2.1 | The First Law

Given the orbital setup in Figure A.1, the distance vector $\vec{r}$ between two stars in a binary is

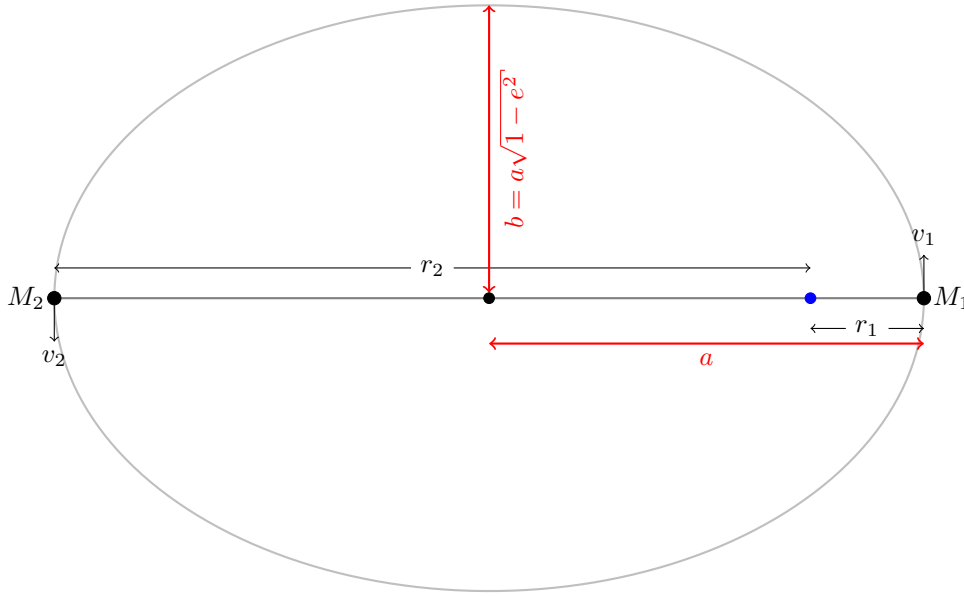


Figure A.1: Figure 2.8 again; A schematic of a simple binary star system with eccentricity e , semimajor axis a , and semiminor axis b . The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a smaller distance r_1 to the barycentre by Kepler's first and Newton's second laws. The elliptical orbital trajectory is shown in grey. Figure adapted from [Richards \(2021\)](#).

$$\vec{r} = \vec{r}_2 - \vec{r}_1, \quad (\text{A.13})$$

with the reduced mass

$$\mu = \frac{M_1 M_2}{M_1 + M_2}, \quad (\text{A.14})$$

and total mass

$$M = M_1 + M_2. \quad (\text{A.15})$$

Since the angular momentum vector is given by

$$\vec{L} = \vec{r} \times \vec{p}, \quad (\text{A.16})$$

its derivative is

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} = 0, \quad (\text{A.17})$$

where the equality follows as $\vec{v} \equiv \frac{d\vec{r}}{dt}$ is parallel to $\vec{p}$, and $\vec{r}$ is parallel to $\frac{d\vec{p}}{dt} \equiv \vec{F}$. This is equivalent to a one-body problem where a mass $\mu = \frac{M_1 M_2}{M_1 + M_2}$ orbits a central mass of mass $M_1 + M_2$. The angular momentum is

$$\vec{L} = \mu \vec{r} \times \vec{v} \quad (\text{A.18})$$

$$\begin{aligned}
&= \mu |\vec{r}| \hat{\mathbf{r}} \times \frac{d}{dt} (|\vec{r}| \hat{\mathbf{r}}) \\
&= \mu |\vec{r}| \hat{\mathbf{r}} \times \left(\frac{dr}{dt} \hat{\mathbf{r}} + r \frac{d\hat{\mathbf{r}}}{dt} \right) \\
&= -\frac{dr}{dt} \hat{\mathbf{r}} \times \mu |\vec{r}| \hat{\mathbf{r}} - r \frac{d\hat{\mathbf{r}}}{dt} \times \mu |\vec{r}| \hat{\mathbf{r}} \\
&= -\frac{dr}{dt} \mu |\vec{r}| (\hat{\mathbf{r}} \times \hat{\mathbf{r}}) - |\vec{r}|^2 \mu \left(\frac{d\hat{\mathbf{r}}}{dt} \times \hat{\mathbf{r}} \right) \\
&= -|\vec{r}|^2 \mu \left(\frac{d\hat{\mathbf{r}}}{dt} \times \hat{\mathbf{r}} \right) \\
&= |\vec{r}|^2 \mu \left(\hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt} \right) \\
&= \mu |\vec{r}|^2 \hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt}. \quad (\text{A.19})
\end{aligned}$$

We seek an expression for $|\vec{r}|$ to determine the trajectory of the stars. We write

$$\begin{aligned}
\vec{a} \times \vec{L} &= -\frac{GM}{|\vec{r}|^2} \hat{\mathbf{r}} \times \left(\mu |\vec{r}|^2 \hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt} \right) \\
&= GM\mu \hat{\mathbf{r}} \times \left(\hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt} \right) \\
&= -GM\mu \left[\left(\hat{\mathbf{r}} \cdot \frac{d\hat{\mathbf{r}}}{dt} \right) \hat{\mathbf{r}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) \frac{d\hat{\mathbf{r}}}{dt} \right] \\
&= GM\mu \frac{d\hat{\mathbf{r}}}{dt}.
\end{aligned}$$

Now consider

$$\begin{aligned}
\frac{d}{dt} (\vec{v} \times \vec{L}) &= \frac{d\vec{v}}{dt} \times \vec{L} + \vec{v} \times \frac{d\vec{L}}{dt} \\
&= \vec{a} \times \vec{L} \\
&= \frac{d}{dt} (GM\mu \hat{\mathbf{r}}), \quad (\text{A.20})
\end{aligned}$$

where $\frac{d\vec{L}}{dt} = 0$ by Eqn(A.17). Hence

$$\vec{v} \times \vec{L} = GM\mu \hat{\mathbf{r}} + \vec{D}, \quad (\text{A.21})$$

where $\vec{D}$ is a constant of integration that exists in the plane of the orbit. We can then introduce the vector $\vec{r}$ into this equation:

$$\begin{aligned}\vec{r} \cdot (\vec{v} \times \vec{L}) &= GM\mu |\vec{r}| \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} + \vec{r} \cdot \vec{D} \\ (\vec{r} \times \vec{v}) \cdot \vec{L} &= GM\mu r + rD \cos \theta.\end{aligned}\tag{A.22}$$

Now by Eqn(A.19), this is simply

$$\frac{|\vec{L}|^2}{\mu} = GM\mu |\vec{r}| \left(1 + \frac{|\vec{D}| \cos \theta}{GM\mu} \right).\tag{A.23}$$

We define the eccentricity of the ellipse as

$$e = \frac{|\vec{r}|}{GM\mu}\tag{A.24}$$

and hence we have

Kepler's First Law

$$|\vec{r}| = \frac{|\vec{L}|^2 / \mu^2}{GM (1 + e \cos \theta)},\tag{A.25}$$

which is the equation defining an ellipse, provided that¹ $e < 1$ – thus proving the First Law. Since the general equation for an ellipse with semimajor axis a is

$$|\vec{r}| = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad \text{for } 0 \leq e < 1,\tag{A.26}$$

it is immediately apparent that we can write

$$|\vec{L}| = \mu \sqrt{GMa(1 - e^2)},\tag{A.27}$$

and since $|\vec{L}|$ is constant by Eqn(A.17), presuming total mass M remains constant we have that μ is constant as well.

A.2.2 | The Second Law

Consider the area of an elliptical orbit (see the shaded area in Figure A.2). We have

$$dA = dr (r d\theta) = r dr d\theta,\tag{A.28}$$

and integrating over r gives

¹Or equivalently that $|\vec{D}| < GM\mu$, which we are free to select as required as the vector $\vec{D}$ is arbitrary.

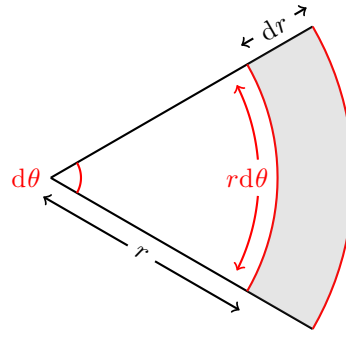


Figure A.2: An infinitesimal area segment of an orbital trajectory. Given the definitions in the figure, we have the area of the shaded segment is $dA = dr(rd\theta) = r dr d\theta$.

$$dA = \frac{1}{2} r^2 d\theta, \quad (\text{A.29})$$

thus

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}. \quad (\text{A.30})$$

We can write the velocity of the star in the plane of the ellipse in terms of its radial component $\vec{v}_r$ and its angular component $\vec{v}_\theta$,

$$\vec{v} = \vec{v}_r + \vec{v}_\theta = \frac{dr}{dt} \hat{r} + |\vec{r}| \frac{d\theta}{dt} \quad (\text{A.31})$$

Importantly we have

$$v_\theta = r \frac{d\theta}{dt}, \quad (\text{A.32})$$

such that

$$\frac{dA}{dt} = \frac{1}{2} r v_\theta. \quad (\text{A.33})$$

Since r and v_θ are orthogonal, we have

$$r v_\theta = \underbrace{|\vec{r} \times \vec{\theta}|}_{\text{by Eqn(A.18)}} = \left| \frac{\vec{L}}{\mu} \right| = \frac{L}{\mu}, \quad (\text{A.34})$$

hence

Kepler's Second Law

$$\frac{dA}{dt} = \frac{1}{2} \frac{L}{\mu}. \quad (\text{A.35})$$

By Eqn(A.17), L is constant (and by Eqn(A.27), μ is constant as well), hence we have that $\frac{dA}{dt}$ is constant

over the orbit, and Kepler's second law is established.

A.2.3 | The Third Law

Let the period of the binary star system be given by T . The area of the ellipse swept out by the binary over its orbit is

$$A = \int_0^T \frac{dA}{dt} dt = \frac{1}{2} \frac{L}{\mu} \int_0^T dt = \frac{1}{2} \frac{L}{\mu} T, \quad (\text{A.36})$$

and since the orbit is elliptical, this is equal to πab , and thus

$$\pi ab = \frac{1}{2} \frac{L}{\mu} T, \quad (\text{A.37})$$

or, equivalently,

$$T^2 = \frac{4\pi^2 a^2 b^2 \mu^2}{L^2}. \quad (\text{A.38})$$

Further, from the geometry of an ellipse we have

$$b^2 = a^2 (1 - e^2). \quad (\text{A.39})$$

Using this and Eqn(A.27) in tandem, we have

Kepler's Third Law

$$T^2 = \frac{4\pi^2}{GM} a^3, \quad (\text{A.40})$$

which proves Kepler's third law.

A.3 | Equations of Motion of decaying binaries

The following derivation is adapted from the derivation presented in [Richards \(2021\)](#). In this section we use a positive-definite metric signature ($\text{Tr } \boldsymbol{\eta} = +2$), and the Einstein summation convention where repeated indices are summed shall apply. Further, Greek indices span from 0 to 3, and Latin indices span from 1 to 3.

We have two goals. The first is to derive expressions for the coupled equations $\frac{da}{dt}$ and $\frac{de}{dt}$ describing the decay of a binary due to gravitational radiation. The second is to determine an expression for $T_c(a_0, e_0)$, the coalescence time, which is the time-scale for this to occur.

A.3.1 | $\frac{da}{dt}$ and $\frac{de}{dt}$

In the weak-field limit, the spacetime metric is presumed to be a perturbation of a locally Minkowski metric, that is, it is given by

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}, \quad (\text{A.41})$$

with η the Minkowski metric, $|h| \ll 1$, and $\kappa^2 = 32\pi G$. This metric induces (see, among others, [Einstein, 1916](#); [Peters, 1964](#)) the wave equation

$$\square \bar{h}_{\mu\nu} = -\frac{1}{2}\kappa T_{\mu\nu}, \quad (\text{A.42})$$

where $\mathbf{T}$ is the stress-energy tensor. Given some plane wave solution to this wave equation, of the form

$$h_{\mu\nu} = A e_{\mu\nu} \cos(\vec{k} \cdot \vec{x} - \omega t), \quad (\text{A.43})$$

where $\mathbf{e}$ is a unit polarization tensor, we have the energy density

$$U = \frac{1}{2}c^2\omega^2 A^2. \quad (\text{A.44})$$

The two possible non-zero spacelike and transverse polarizations for a wave travelling (without loss of generality) in the z -direction are

$$e_1 = \frac{1}{\sqrt{2}}(\hat{x}\hat{x} - \hat{y}\hat{y}), \quad (\text{A.45})$$

and

$$e_2 = \frac{1}{\sqrt{2}}(\hat{x}\hat{y} - \hat{y}\hat{x}), \quad (\text{A.46})$$

where we denote the “plus” polarisation as e_1 and the “cross” polarisation as e_2 . The power P radiated into a solid angle Ω is given by

$$\frac{dP}{d\Omega} = \frac{G}{8\pi c^5} \left(\frac{\partial^3 Q_{ij}}{\partial t^3} e_{ij} \right)^2, \quad (\text{A.47})$$

where the components of the mass tensor $\mathbf{Q}$ are given by

$$Q_{ij} = m^\alpha x_{\alpha i} x_{\alpha j}. \quad (\text{A.48})$$

The total power is obtained reasonably straightforwardly:

$$\frac{dP}{d\Omega} = \frac{G}{8\pi c^5} \left[\left(\frac{d^3 Q_{ij}}{dt^3} \right)^2 - 2n_i \frac{d^3 Q_{ij}}{dt^3} n_k \frac{d^3 Q_{kj}}{dt^3} - \frac{1}{2} \left(\frac{d^3 Q_{ii}}{dt^3} \right)^2 + \frac{1}{2} \left(n_i n_j \frac{d^3 Q_{ij}}{dt^3} \right)^2 + \frac{d^3 Q_{ii}}{dt^3} n_j n_k \frac{d^3 Q_{jk}}{dt^3} \right], \quad (\text{A.49})$$

with $\hat{n}$ a unit vector in the direction of radiation. The total power is then found by integrating this equation over all directions, viz.

$$P = \frac{G}{5c^5} \left(\frac{d^3 Q_{ij}}{dt^3} \frac{d^3 Q_{ij}}{dt^3} - \frac{1}{3} \frac{d^3 Q_{ii}}{dt^3} \frac{d^3 Q_{jj}}{dt^3} \right). \quad (\text{A.50})$$

Let us now inject the physics. Suppose the primary star has mass M_1 and the secondary mass M_2 . In the xy -plane, with the origin at the center of mass, we can parameterise the positions of the two stars as

$$\vec{x}_1 = \begin{bmatrix} d_1 \cos \varphi \\ d_1 \sin \varphi \end{bmatrix} \quad \vec{x}_2 = \begin{bmatrix} -d_2 \cos \varphi \\ -d_2 \sin \varphi \end{bmatrix}, \quad (\text{A.51})$$

with

$$d_1 = \frac{m}{m+M} d \quad \text{and} \quad d_2 = \frac{M}{m+M} d, \quad (\text{A.52})$$

and φ our angle parameter. The components of the tensor $\mathbf{Q}$ are

$$Q_{xx} = \mu d^2 \cos^2 \varphi, \quad (\text{A.53})$$

$$Q_{yy} = \mu d^2 \sin^2 \varphi, \quad (\text{A.54})$$

$$Q_{xy} = Q_{yx} = \mu d^2 \sin \varphi \cos \varphi, \quad (\text{A.55})$$

with the rest vanishing, and with μ as the reduced mass. Now, as binary stars obey Kepler's laws, Kepler's orbit equation (Eqn(A.25)) holds:

$$d = \frac{a(1-e^2)}{1+e \cos \dot{\varphi}}, \quad (\text{A.56})$$

whilst the angular velocity is given by

$$\dot{\varphi} = \frac{\sqrt{G(M_1 + M_2)(1-e^2)a}}{d^2}. \quad (\text{A.57})$$

We therefore have

$$\frac{d^3 Q_{xx}}{dt^3} = \gamma(1 + e \cos \dot{\varphi})^2 (2 \sin(2\dot{\varphi}) + 3e \sin \dot{\varphi} \cos^2 \dot{\varphi}), \quad (\text{A.58})$$

$$\frac{d^3 Q_{yy}}{dt^3} = -\gamma(1 + e \cos \dot{\varphi})^2 (2 \sin(2\dot{\varphi}) + e \sin \dot{\varphi} (1 + 3 \cos^2 \dot{\varphi})), \quad (\text{A.59})$$

$$\frac{d^3 Q_{xy}}{dt^3} = \frac{d^3 Q_{yx}}{dt^3} = -\gamma(1 + e \cos \dot{\varphi})^2 (2 \cos(2\dot{\varphi}) - e \cos \dot{\varphi} (1 - 3 \cos^2 \dot{\varphi})), \quad (\text{A.60})$$

where the quantity γ is defined as

$$\gamma^2 \equiv \frac{4G^3 M_1^2 M_2^2 (M_1 + M_2)}{a^5 (1 - e^2)^5}. \quad (\text{A.61})$$

The total power is therefore

$$P = \frac{8}{15} \frac{G^4}{c^5} \frac{M_1^2 M_2^2 (M_1 + M_2)}{a(1 - e^2)^5} (1 + e \cos \varphi)^4 (12(1 + e \cos \varphi)^2 + e^2 \sin^2 \varphi). \quad (\text{A.62})$$

Averaging this over one orbit yields the average power, which we write now as $\langle \frac{dE}{dt} \rangle$,

$$\left\langle \frac{dE}{dt} \right\rangle = -\frac{32}{5} \frac{G^4}{c^5} \frac{M_1^2 M_2^2 (M_1 + M_2)}{a^5 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right). \quad (\text{A.63})$$

The average angular momentum is

$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{32}{5} \frac{G^{\frac{7}{2}} M_1^2 M_2^2 \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right). \quad (\text{A.64})$$

We now write the gravitational energy from the two-body problem,

$$E = -\frac{GM_1 M_2}{2a} \quad (\text{A.65})$$

$$\implies \frac{dE}{dt} = \frac{GM_1 M_2}{2} \frac{1}{a^2} \frac{da}{dt}. \quad (\text{A.66})$$

Define the constant β as

$$\beta \equiv \frac{64}{5} \frac{G^3 M_1 M_2 (M_1 + M_2)}{c^5}. \quad (\text{A.67})$$

Now insert Eqn(A.66) into Eqn(A.63) to obtain

$$\frac{da}{dt} = -\frac{2a^2}{GM_1 M_2} \frac{32}{5} \frac{G^4 M_1^2 M_2^2 (M_1 + M_2)}{c^5 a^5 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (\text{A.68})$$

$$= -\frac{\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right), \quad (\text{A.69})$$

which is our first equation. The derivation for $\frac{de}{dt}$ is more involved. The angular momentum from the

two-body problem is

$$L^2 = \frac{GM_1^2 M_2^2 a (1 - e^2)}{M_1 + M_2} \quad (\text{A.70})$$

Implicit differentiation reveals that

$$2L \frac{dL}{dt} = \frac{GM_1^2 M_2^2}{M_1 + M_2} \frac{d}{dt} (a (1 - e^2)) \quad (\text{A.71})$$

$$= \frac{GM_1^2 M_2^2}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right]. \quad (\text{A.72})$$

Multiplying each side of Eqn(A.64) by $2L$ and substituting this in, we obtain

$$\begin{aligned} \frac{GM_1^2 M_2^2}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right] &= -L \frac{64 G^{\frac{7}{2}} M_1^2 M_2^2 \sqrt{M_1 + M_2}}{5 c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right) \\ \frac{1}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right] &= -\sqrt{\frac{GM_2^2 M_1^2 a (1 - e^2)}{M_1 + M_2}} \frac{64 G^{\frac{5}{2}} \sqrt{M_1 + M_2}}{5 c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right) \\ \frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} &= -\frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8} e^2 \right) \\ 2ae \frac{de}{dt} &= \frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8} e^2 \right) + \frac{da}{dt} (1 - e^2) \\ 2ae \frac{de}{dt} &= \frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8} e^2 \right) - \frac{\beta}{a^3 (1 - e^2)^{\frac{5}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \\ \frac{de}{dt} &= \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \frac{(1 - e^2) \left(1 + \frac{7}{8} e^2 \right) - \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right)}{2e} \\ &= \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left[-\frac{19}{12} e^3 - \frac{121}{192} e \right] \\ &= -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right), \end{aligned} \quad (\text{A.73})$$

which is our second equation. Hence we arrive at the coupled differential equations describing the decay of a binary star system:

Peters (1964) equations for the inspiral of two compact objects

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (\text{A.74a})$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right) \quad (\text{A.74b})$$

A.3.2 | The coalescence time

The coalescence time is by definition the time it takes for the binary system to merge after the secondary supernova. We can step towards this in a similar way to [Peters \(1964\)](#). First, dividing Eqn(A.74a) by Eqn(A.74b) gives

$$\frac{da}{de} = \frac{12}{19} \frac{a}{e} \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{(1 - e^2) \left(1 + \frac{121}{304}e^2\right)} \quad (\text{A.75})$$

We can integrate this however we choose to obtain $a(e)$, namely

$$a(e) = \frac{c_0 e^{\frac{12}{19}}}{(1 - e^2)} \left(1 + \frac{121}{304}e^2\right)^{\frac{870}{2299}} \quad (\text{A.76})$$

with

$$c_0 \equiv a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304}e_0^2\right)^{-\frac{870}{2299}} \quad (\text{A.77})$$

embodying the initial parameters. Now, substituting Eqn(A.76) into Eqn(A.74b), we see

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3\right) \quad (\text{A.78})$$

$$= -\frac{19}{12} \frac{\beta}{\left(\frac{c_0 e^{\frac{12}{19}}}{(1 - e^2)} \left(1 + \frac{121}{304}e^2\right)^{\frac{870}{2299}}\right)^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3\right) \quad (\text{A.79})$$

$$= -\frac{19}{12} \frac{\beta (1 - e^2)^{\frac{3}{2}} e}{c_0^4 e^{\frac{48}{19}} \left(1 + \frac{121}{304}e^2\right)^{\frac{3480}{2299}}} \left(1 + \frac{121}{304}e^2\right) \quad (\text{A.80})$$

$$= -\frac{19}{12} \frac{\beta (1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}}{c_0^4 \left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}} \quad (\text{A.81})$$

$$(\text{A.82})$$

Now write this as $\frac{dt}{de}$ and integrate to get the coalescence time:

$$\frac{dt}{de} = -\frac{12}{19} \frac{c_0^4 \left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{\beta (1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} \quad (\text{A.83})$$

$$T_c = \int_0^{T_c} dt = -\frac{12}{19} \frac{c_0^4}{\beta} \int_{e_0}^0 \frac{\left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} de \quad (\text{A.84})$$

$$= \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{\left(1 + \frac{121}{304}e^2\right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} de \quad (\text{A.85})$$

i.e.,

The coalescence time of inspiralling compact objects

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de \quad (\text{A.86})$$

If $e_0 = 0$ then the orbit is circular, and we immediately see concerning behaviour – the integral collapses to zero, necessitating a different approach.

One of the critical mistakes here is presuming that c_0 in Eqn(A.77) is valid for all physical eccentricities. In fact, in the limit as $e_0 \rightarrow 0$, c_0 diverges. We retreat to before this critical assumption, and examine Eqn(A.74a). In the limiting case of a circular orbit, $e = e_0 = 0$, so the expression becomes

$$\frac{da}{dt} = -\frac{\beta}{a^3}. \quad (\text{A.87})$$

and by separation of variables we obtain

$$a^3 da = -\beta dt, \quad (\text{A.88})$$

of which the solution is

$$\frac{a^4}{4} = -\beta t + c, \quad (\text{A.89})$$

for constant $c \in \mathbb{R}$. The boundary conditions are that when $t = 0, a = a_0$, and when $t = T_c, a = 0$. Applying the first boundary condition, we have $c = a_0^4$. Hence,

$$a(t) = (a_0^4 - 4\beta t)^{\frac{1}{4}}, \quad (\text{A.90})$$

and imposing our second boundary condition we obtain the coalescence time

$$T_c = \frac{a_0^4}{4\beta}. \quad (\text{A.91})$$

But if $e \neq 0$, then the orbit is eccentric and the integral must be solved numerically, as no closed form solution exists.

A.4 | Construction of the wind mass-loss rate

We construct our wind mass loss rates as follows. The total pre-WR mass loss is given by [de Jager et al. \(1988\)](#), namely

$$\log \dot{M}_{\text{de Jager}} = - \sum_{n=0}^6 \sum_{\substack{i=0 \\ j=n-i}}^n a_{ij} \cos(i \arccos x_1) \cos(j \arccos x_2), \quad (\text{A.92})$$

$$\dot{M}_{\text{pre-WR}} = \sqrt{\frac{Z}{Z_{\odot}}} 10^{\log \dot{M}_{\text{de Jager}}}, \quad (\text{A.93})$$

where $x_1 = \frac{\log T - 4.05}{0.75}$, $x_2 = \frac{\log L - 4.6}{2.1}$, and the coefficients a_{ij} are given by the entries in the matrix

$$\mathbf{a}_{ij} = \begin{pmatrix} 6.34916 & -5.04240 & -0.83426 & -1.13925 & -0.12202 \\ 3.41678 & 0.15629 & 2.96244 & 0.33659 & 0.57576 \\ -1.08683 & 0.41952 & -1.37272 & -1.07493 & - \\ 0.13095 & -0.09825 & 0.13025 & - & - \\ 0.22427 & 0.46591 & - & - & - \\ 0.11968 & - & - & - & - \end{pmatrix}. \quad (\text{A.94})$$

This mass loss rate is based upon the rates for 271 stars, the majority of which are Population I. Individual terms in the summation are Chebyshev polynomials of the first kind, restricted over the domain $\log T_{\text{eff}} \in [3.3, 4.8]$, $\log L \in [2.5, 6.7]$. As the free parameters here are temperature and luminosity, the result can be plotted over a HRD. The results of this are shown in Figure A.3. Generally we constrain the inputs to the arccosine function to be between $-1 \leq x \leq 1$, which permits them to be defined outside of the defined area from [de Jager et al. \(1988\)](#); the effect of this is shown in the right-hand panel of Figure A.3. Without this restriction, the mass loss rates are simply undefined there, and are thus set to zero within the code.

We interpolate the bi-stability jump around 25 kK (see, for example, [Pauldrach & Puls, 1990](#); [Vink et al., 1999](#)) with the following regime. Define the constants

$$C_1 = \left(\frac{Z}{Z_{\odot}} \right)^{0.13}, \quad (\text{A.95})$$

$$C_2 = -c_1 + c_2 \log \left(\frac{L}{c_{16}} \right) - c_3 \log \left(\frac{M}{c_{15}} \right) - c_4 \log (1.3C_1) + c_5 \log \left(\frac{T}{c_{14}} \right) - c_6 \left(\log \left[\frac{T}{c_{14}} \right] \right)^2 + c_7 \log \left(\frac{Z}{Z_{\odot}} \right), \quad (\text{A.96})$$

and

$$C_3 = -c_8 + c_9 \log \left(\frac{L}{c_{16}} \right) - c_{10} \log \left(\frac{M}{c_{15}} \right) - c_{11} \log (0.65C_1) + c_{12} \log \left(\frac{T}{2 \times 10^4} \right) + c_{13} \log \left(\frac{Z}{Z_{\odot}} \right), \quad (\text{A.97})$$

where the constants c_i are given in Table A.1.

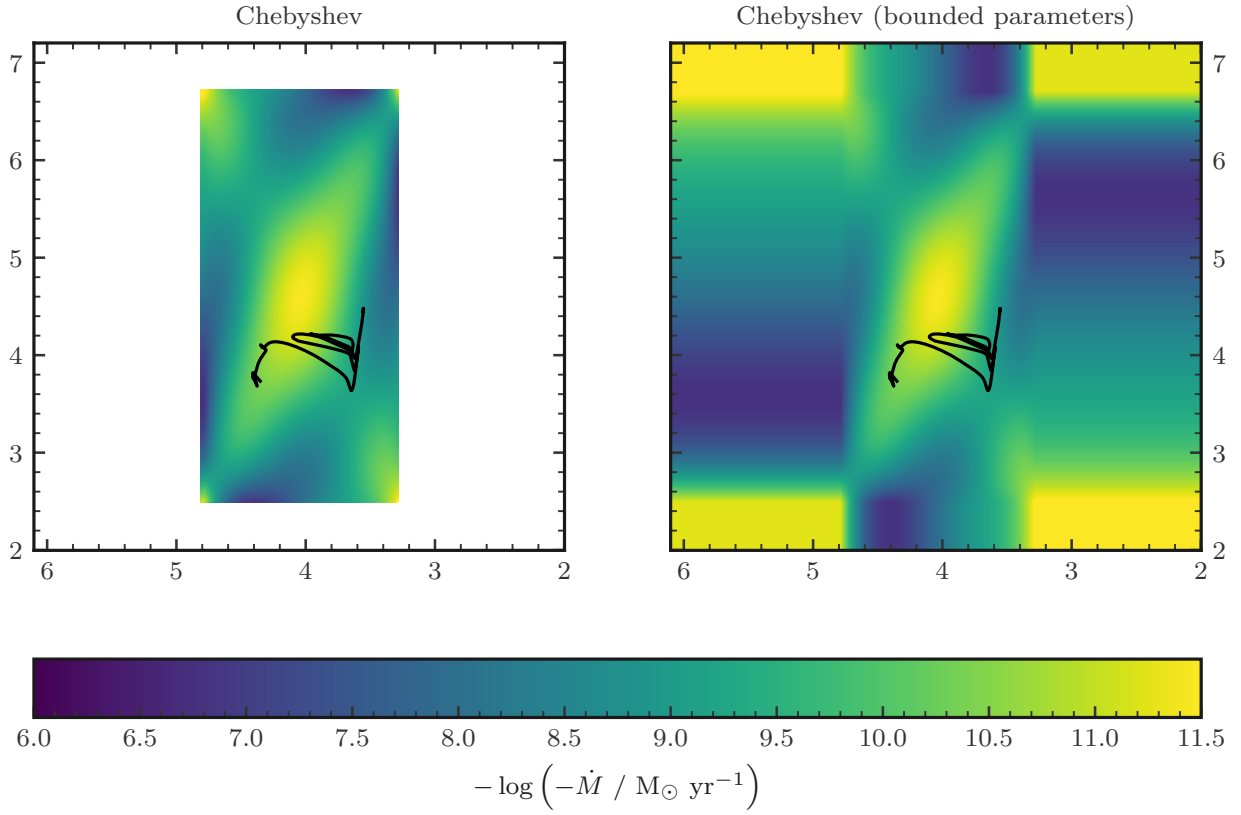


Figure A.3: The Chebyshev polynomial expansion of Eqn(A.92). A $10 M_{\odot}$ model is overplotted in black for reference. The “bounded parameters” plot restricts the arguments to arccos in Eqn(A.92) to be between $-1 \leq x_i \leq 1$.

Constant	Value	Constant	Value	Constant	Value	Constant	Value
c_1	6.697	c_5	0.933	c_9	2.210	c_{13}	0.85
c_2	2.194	c_6	10.92	c_{10}	1.339	c_{14}	4×10^4 K
c_3	1.313	c_7	0.85	c_{11}	1.601	c_{15}	$30 M_{\odot}$
c_4	1.226	c_8	6.688	c_{12}	1.07	c_{16}	$10^5 L_{\odot}$

Table A.1: The mass loss rate constants for interpolation around the bi-stability jump (Eqns (A.95–97)). This interpolation is shown pictorially in Figure A.4.

Our mass loss rate becomes:

$$\dot{M}_{\text{BS}} = \begin{cases} \frac{(T-T_1)10^{C_3} + (T_2-T)\dot{M}_{\text{pre-WR}}}{2.5 \times 10^3} & T_1 < T < T_2 \\ 10^{C_2} & T_2 \leq T \leq T_3 \\ \frac{(T-T_3)10^{C_2} + (T_4-T)10^{C_3}}{5 \times 10^3} & T_3 < T < T_4 \\ \frac{(6T_1-T)10^{C_2} + (T-5T_1)\dot{M}_{\text{pre-WR}}}{10^4} & 5T_1 < T < 6T_1 \\ \dot{M}_{\text{pre-WR}} & \text{otherwise} \end{cases} \quad (\text{A.98})$$

where $T_1 = 10 \text{ kK}$, $T_2 = 12.5 \text{ kK}$, $T_3 = 22.5 \text{ kK}$, $T_4 = 27.5 \text{ kK}$. This interpolation is shown in Figure A.4.

We define

$$\mathcal{M} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-13.6 + 1.63 \log\left(\frac{L}{L_\odot}\right) + 2.22 \log X}, \quad (\text{A.99})$$

where X is the surface hydrogen abundance of the primary, and then define

$$\dot{M}_{\text{WNL}} = \begin{cases} 5 \left(\mathcal{M} - \dot{M}_{\text{BS}} \right) (0.42 - X) + \dot{M}_{\text{BS}} & 0.42 > X > 0.4 \\ \mathcal{M} & \text{otherwise} \end{cases}. \quad (\text{A.100})$$

Our total mass loss (provided the star has not depleted its surface helium, which will be discussed below) is now

$$\dot{M}_{\text{A}} = \begin{cases} 10\dot{\mu} (\log T - 3.9) + \dot{M}_{\text{BS}} & X < 0.42 \text{ and } \log T > 3.9 \\ \dot{M}_{\text{WNL}} & X > 0.42 \text{ and } \log T > 4.0 \\ \dot{M}_{\text{BS}} & \text{otherwise} \end{cases}, \quad (\text{A.101})$$

where $\dot{\mu} = \dot{M}_{\text{WNL}} - \dot{M}_{\text{BS}}$. Finally, if the surface helium depletes we modify the mass loss rate further, defining

$$\dot{M}_{\text{WC}} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-8.3 + 0.84 \log\left(\frac{L}{L_\odot}\right) + 2.04 \log X_{\text{He}} + 1.04 \log(1 - X_{\text{He}})} \quad (\text{A.102})$$

such that our total mass loss rate becomes (provided $X_{\text{H}} < 10^{-3}$ and $\log T > 4$)

$$\dot{M}_{\text{B}} = \begin{cases} 100 \left(\dot{M}_{\text{WC}} - \dot{M}_{\text{WNL}} \right) (\mathcal{X} - 0.02) + \dot{M}_{\text{WNL}} & \mathcal{X} > 0.02 \\ \dot{M}_{\text{WNL}} & \mathcal{X} < 0.02 \end{cases}, \quad (\text{A.103})$$

where $\mathcal{X} = \frac{\frac{1}{3}X_{\text{C}} + \frac{1}{4}X_{\text{O}}}{X_{\text{He}}}$. The total wind mass-loss rate is then

$$\dot{M}_{\text{wind}} = \begin{cases} \dot{M}_{\text{B}} & \text{surface helium depleted} \\ \dot{M}_{\text{A}} & \text{otherwise} \end{cases}. \quad (\text{A.104})$$

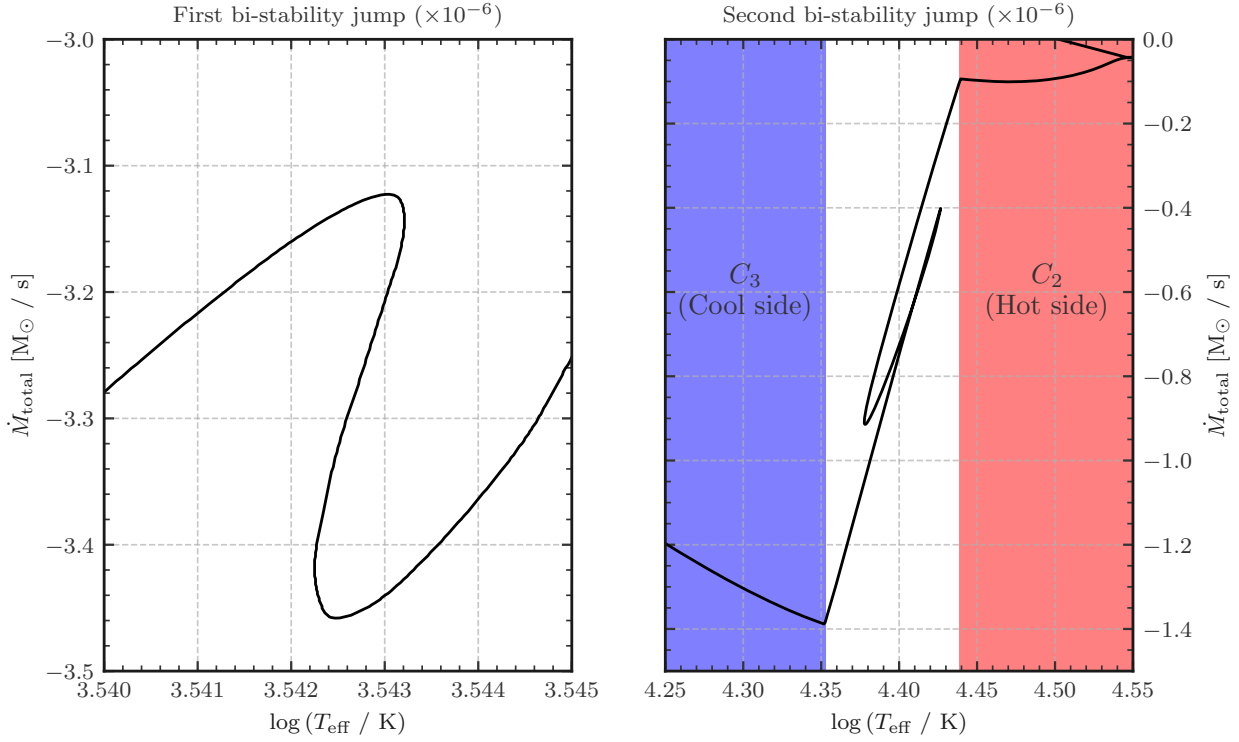


Figure A.4: An illustration of the bi-stability jumps in the mass-loss prescription. This illustration was computed for a $21 \text{ M}_{\odot}$ star. *Left plot:* The lower bi-stability jump, *Right plot:* The upper bi-stability jump. Shown in blue is the cool side of the bi-stability jump, where C_3 (Eqn(A.97)) is the interpolation, and shown in red is the hot side, where C_2 (Eqn(A.96)) is the interpolation.